Hao-Yi (Karen) Tu

karenhaoyitu@gmail.com ♦ (346) 368-4758
linkedin.com/in/karenhytu ♦ karenhaoyitu.com ♦ github.com/haoyt5

Professional Experience

ByteDance Inc. San Jose, CA
Full-Stack Software Engineer *September 2023 – June 2026*

- Built a global security operations workspace using **React, TypeScript, and Go (Kitex RPC)**, supporting **50+ users** and **10+ cross-region workflows**.
- Independently developed **Go API/RPC endpoints and backend business logic** to integrate alerts, threat intelligence, and sandbox results from **7 sources** into a unified investigation workflow.
- Embedded **AI-assisted case summaries** and decision-node automation into existing workflows, reducing tool switching and reaching **5 of 6 analyst teams** within two quarters.
- Built event-triggered AI report regeneration and **Server-Sent Events (SSE)** streaming to progressively deliver results from long-running AI operations.
- Diagnosed and resolved cross-region production incidents using **logs and metrics**, reducing recurring service disruptions for **24/7 global security operations**.
- Led **React architecture reviews**, mentored engineers, and improved release quality through proactive code reviews.

Johnson Controls Inc. Campbell, CA
Software Engineering Intern *June 2022 – August 2022*

- Built a full-stack Android app distribution portal for smart-home panels using **TypeScript, React, Express.js, and PostgreSQL**.
- Developed **10+ REST APIs** for version management, validation, and uploads; used **Sequelize, Ajv, and Docker** to support reliable development and testing.
- Improved backend maintainability using **Sequelize ORM** and **Ajv JSON-schema validation**, refining data integrity and API reliability.
- Containerized client, API, auth, and DB services with **Docker** to streamline local development and testing.

Trend Micro Inc. Taipei, Taiwan
Sr. Software Engineer *June 2019 – August 2021*

- Developed **React/Redux** interfaces for activation and credit-based licensing services, enabling **7,000+ enterpricse customers** to activate products and track license usage.
- Migrated a multi-page application to a single-page architecture, increasing activation success by **120%** in four months.
- Led development of a user behavior analytics dashboard, reducing manual reporting effort by **60%+** for business stakeholders.
- Built a channel sales analytics platform with data visualization and **role-based access control (RBAC)** for European and North American teams.
- Reduced **first contentful paint** time by **50%** using paginated loading and caching optimizations.
- Won **2nd place among 90 teams** and 1,500 employees in an internal Customer Success and DevOps contest.
- Won **1st place among 20 teams** in the Information System (IS) DevOps Camp.

Skills

Languages: TypeScript, JavaScript (ES6+), Go
Frontend: React, Redux, HTML, CSS
Backend: Node.js, Express.js, REST APIs, RPC (Kitex), GraphQL, Server-Sent Events (SSE)
Databases: PostgreSQL, MongoDB
Tools: Docker, Git, Jest, React Testing Library, CI/CD, monitoring

Education

Carnegie Mellon University Mountain View, CA
Master of Science in Software Engineering *January 2022 – May 2023*
- Selected Coursework: Software Design and Architecture, Browser Security

New York University, Tandon School of Engineering Brooklyn, NY
Preparatory Course for Graduate Studies in Computing (NYU Tandon Bridge) *January 2021 – July 2021*

National Taiwan Normal University Taipei, Taiwan
Bachelor of Arts, History *September 2012 – January 2017*